



B-29

GOING NUCLEAR

A GLOBAL WAR 3D PRINTED EXPANSION SET



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A Global War 3rd Edition 3d printed Expansion Set

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Overview (v3.0)



Welcome to "GOING NUCLEAR", Historical Board Gaming's nuclear war expansion for Global War. This is the first of many 3D printed piece sets to supplement HBG's Global War game. All of our 3D printed rule sets are denoted by the letters TDP with a letter after them to indicate the set (TDPA, TDPB etc....) for easy reference in future rule sets.

In this expansion you're going to learn how to build an atomic bomb. You and your opponents will be scavenging the globe for resources and constructing facilities in order to "GO NUCLEAR!" Also included in this set are rules to help Axis & Allies 1940 & Global 1939 players connect these rules to those game systems.



Set Contents

Going Nuclear Base Set	Going Nuclear Extreme Set
• B-29 Bomber (x1) • “Fat Man” Atomic Bomb (x2) • Nuclear Blast Marker (x2) • Research Marker (x7) • Heavy Water Marker (x4) • Uranium Marker (x8) • Nuclear Reactor Marker (x4) • Radiation Marker (x6) • Rule Set (x1)	• B-29 Bombers (x2) • “Fat Man” Atomic Bombs (x3) • Nuclear Blast Marker (x4) • Research Markers (x10) • Heavy Water Markers (x6) • Uranium Markers (x12) • Nuclear Reactor Marker (x8) • Radiation Marker (x8) • Rule Set (x1)

HOW DO YOU BUILD AN ATOMIC BOMB?

How do you build an atomic bomb? No question burned heavier in the minds of physicists and world leaders in the 1940s. A simple version of the process works something like this:

First, you need the best scientific minds in the world and facilities for them to use. The **Research** marker represents these.

Second you need a source of **Uranium**. As of 1940 there were only four places in the world you could find it: Colorado, the Belgian Congo, Northern Canada, and Czechoslovakia.

Third, you need a **Nuclear Reactor** to enrich the uranium so that will be usable in an atomic bomb.

Finally you need **Heavy Water**. Heavy water contains a larger than normal amount of the hydrogen isotope *deuterium*. Heavy water acts as a moderating element in sustaining a nuclear reaction as well as a coolant to the reactor. At the start of the war there were only two facilities in the world capable of producing heavy water. These were located in Telemark, Norway and Dnepropetrovsk, USSR.



Figure : Vermark Hydroelectric Station in Telemark Norway (1935)

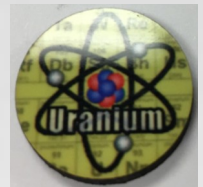


TDPA 1.0: COMPONENTS OF A BOMB

1.1 Research: The **Research** marker represents Scientists and necessary facilities for them to engage in atomic research. The USA, USSR and Germany each begin the game with research markers. Any other major power must purchase them for 15 IPP. The marker must be placed in the home country. If the zone that contains the research marker is captured, the capturing player gains two free rolls on the atomic bomb technology table for each stage the conquered player has acquired. The research marker may not be moved or tactically bombed.



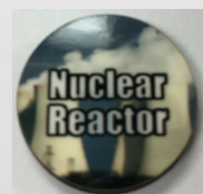
1.2 Uranium: Several territories begin with a **Uranium** marker in them. This marker represents a uranium mine. These are the only sources of uranium available in the game. They may never be destroyed or damaged but may be captured. A uranium mine may not be moved but you may collect uranium from it as outlined below.



1.3 Heavy Water: Two territories (Southern Norway & Western Ukraine) begin with a **Heavy Water** marker, denoting the location of a heavy water plant. These are indicated in the set up changes in rule 6.0. A heavy water plant is a facility and can be captured and tactically bombed. It must be undamaged to produce heavy water. A Heavy Water plant can be constructed for 9 IPP and takes 3 turns to build thus a 3/3/3 unit. It can sustain up to 5 points of damage. It has inherent AA guns like a factory.



1.4 Nuclear Reactor: A player must possess a nuclear reactor in order to build an atomic bomb. A Nuclear Reactor is a facility requiring 4 turns to build. It costs 3/3/3 IPP. It may be tactically bombed up to 6 damage points. A nuclear reactor must be undamaged to build an atomic bomb. It has inherent AA guns like a factory. A nuclear reactor can be captured. This gives the capturing player 2 free research rolls towards nuclear technology and half the player's off-board supply of heavy water and uranium (see TDPA 2.2)

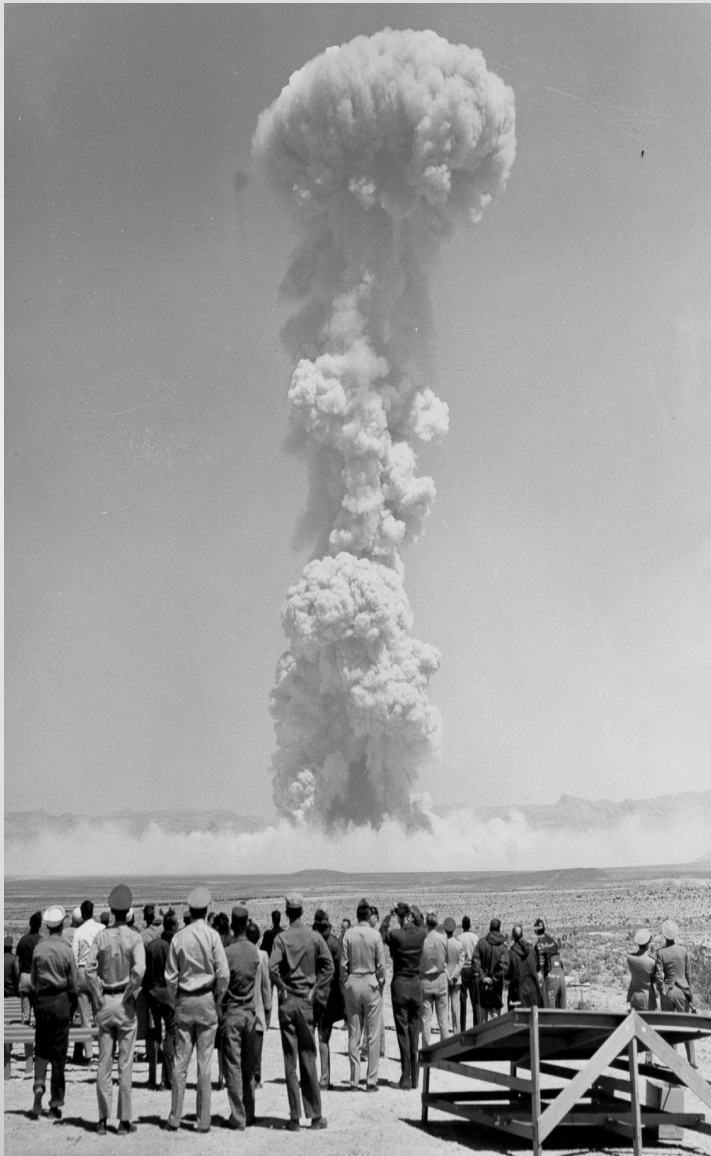




TDPA 2.0 BUILDING THE BOMB

2.1 Assembling Components: In order to build an atomic bomb you must have a functional nuclear reactor and sufficient stocks of heavy water and uranium. You must also have developed atomic bomb technology. No components can be assembled by any player until July of 1939.

2.2 Off-Board Stock Piles: To keep your Uranium Mines and Heavy Water plants from being confused with your stock piles of nuclear material, you will keep resources you acquire off board with your money. Use the heavy water and uranium markers with chips underneath them to keep track.



2.2 Moving Uranium and Heavy

Water: You move Uranium and Heavy Water to yourself (or another member of your Alliance). Each successful delivery gives the receiving player 1 Heavy Water and 1 Uranium marker which the player will add to their off-board supply. Follow the procedure described below:

2.2.1 During the Collect Income and Place Units Phase of your turn you may attempt to move one heavy water and / or one uranium marker from each heavy water plant / uranium mine you possess.

2.2.2 Such movement may cross any number of land zones possessed by your alliance. If such a path crosses sea zones, it must travel via the shortest route towards a major port in the receiving nation.

2.2.3 If such a path moves through a zone containing enemy naval vessels it is subject to interdiction as if it were a lend-leased military unit per Global War rules (i.e. a successful raiding attempt will destroy the resource). Once the resource reaches your home nation it becomes part of your off-board stockpile.

2.3 Developing Technology: Players may conduct atomic research starting July of 1939 so long as they have a research marker [they do not have to be at war with a major power]. Players pay 2 IPP and roll a D12, placing their roundel on each stage after successfully completing it (i.e. rolling 10 or higher).

Atomic Bomb Development Table

ATOMIC BOMB	Stage 1	Stage 2	Stage 3	Complete
	10+	10+	10+	10+

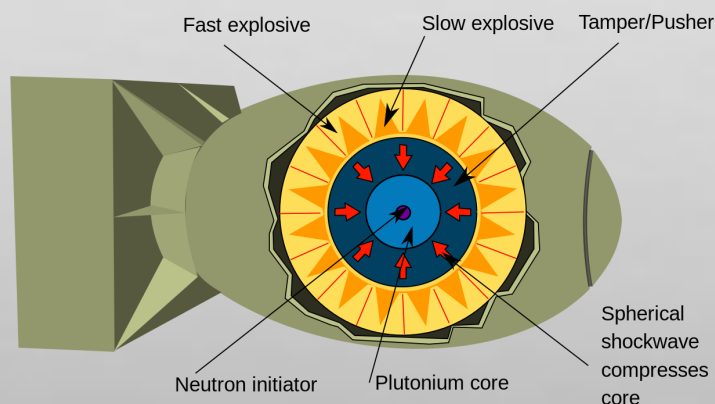
2.4 Building the A Bomb: Once players have completed the required research and have stockpiles of Uranium and Heavy Water and have a Nuclear Reactor they may build an Atomic Bomb as they would any other military unit. Each atomic bomb costs 5IPP to construct and requires the player to expend 3 heavy water and 3 uranium from their acquired supplies.

2.5 The Bomb: You may place your completed bombs on the board at a nuclear reactor during your place units phase. A bomb may be transported in combat movement as outlined below in TDPA 4.0 or in non-combat movement via strategic rail movement, naval transport, air transport or by a strategic bomber.

2.6 Cooperation: Alliance members may cooperate to build an atomic bomb. For example, players may lend-lease each other money for atomic research, deliver strategic materials to one another using the lend-lease rules



TDPA 3.0 DROPPING THE BOMB



The bomb represented in this game is the US "Fat Man" atomic bomb. This was a 10,300lb nuclear bomb with a 21KT yield.

3.1 Delivery: A player delivers an atomic bomb from a strategic bomber to a land territory. A bomber that survives air superiority combat and anti-aircraft fire automatically delivers the bomb to its target. It may not be dropped if there are friendly ground forces in (i.e. attacking) the same zone (excluding Partisans).

3.2 Attacking with the Bomb: Place the Nuclear Blast marker in the attacked zone. The atomic bomb attacks all ground and air units present in a zone. The attacking player rolls 4D12, hitting on a 1-8 with each die.



3.3 Strategic Bombing: An atomic bomb may be used to strategic bomb factories, such an attack does 6D6 strategic damages.

3.4 Tactical Bombing: An atomic bomb can be used to completely destroy any one non-factory facility (air base, naval base etc....)



3.5 Radiation: Place a Radiation marker in the attacked zone at the end of the turn. The radiation marker permanently decreases the value of the territory by 1 until the end of the game or for five years.

3.6 Limitations: Allied nations may only drop an atomic bomb on the home country of a nation they are at war with. Thus, the only viable targets for the Allies, while at war with the Axis would be the home country of Germany, Italy and Japan. The Comintern and Axis do not face such restrictions.





TDPA 4.0 B-29 [Long Range, Heavy Strategic Bomber]

The B-29 Superfortress is an advanced American bomber that can only be built once the player has developed both Long Range Aircraft and Heavy Bombers. The B-29 has a range of over 3200 miles with a 20,000LB payload. It features (12) .50 caliber machine guns in remote controlled turrets for defense.

Unit	Attack	Defense	Move	Cost
B-29	(2)* X 5**	4	10	18

* Intercept value

** Carpet Bomb Value



TDPB 5.0 VICTORY CONDITIONS

All Nations score 1 Victory Condition if you end the game in possession of a uranium mine, a heavy water plant, a nuclear reactor and nuclear technology.





TDPB 6.0 SET UP CHANGES

6.0 Add the following to Global 1936 and 1939 Set-Ups.

Uranium Marker



Great Plains (USA) *Colorado Front Range*

Slovakia *Jachymov*

Belgian Congo (Belgium) *Shinkolobwe*

Ontario (Canada) *Eldorado Mine*



TDPA 7.0 ADVANCED OPTIONS

7.1 Japanese Bomb: Add a research marker in Tokyo and a Heavy Water plant in Korea and Hokkaido.

7.2 Uranium Transport: A nation may transport uranium via submarine. To do so, uranium must move over land via a *supply path* from a uranium mine, to a friendly controlled port, where it then must be loaded onto a submarine if the submarine were a transport. The submarine moves normally. Once the submarine reaches a friendly controlled port in the receiving nation, the uranium is added to the receiving nation's off-board stockpile.



7.3 Carpet Bombing: Carpet bombing is an attack available only to Strategic bombers. Strategic bombers (and heavy strategic bombers) may attack forces in a land zone rolling 3 dice and getting a hit on each dice on a "2" or less (5 dice for heavy bombers). The attacking bomber is subject to air superiority combat and AA fire from AA guns (not facility guns) as if it were strategic bombing. No other land forces may fire back. Combat lasts 1 round after which the bomber must retreat.



APPENDIX A: GLOBAL WAR 1939 CONVERSION

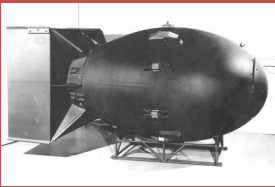
If you want to use these rules in Global War 1939, the following changes will need to be made;

Home Nation: The Home Nation is well defined for players in Global War, however a definition is warranted for players of Global 1939 and Axis & Allies 1940. So for simplicity your home nation is;

Home Nation: Your nation's capitol, and each land zone that is contiguous with it you control at start of game. The home nation will never include conquered territories, colonies, or islands.

2.2 Transporting Uranium and Heavy Water: If you possess a territory with Uranium marker or Heavy Water marker you automatically have access to each component and can stockpile one of each per turn so long as you can trace a path from the marker to your home nation over any number of land and sea zones free of enemy units.

2.3 Research: Players may conduct atomic research starting the first turn of 1939 so long as they have a research marker [they do not have to be at war with a major power]. Players pay 2 IPP and roll a D12, accumulating research points like they do for other technologies. Players can begin their research without having all the components but must have them by stage 3 as shown below.

ATOMIC BOMB	Stage 1	Stage 2	Stage 3	Complete
	20 points	20 points	20 points	20 points

3.2-3.4 Attacking with the Bomb: Ignore reference to air superiority combat (that is a Global War rule).

Unit	Location
Research Markers 	San Francisco (USA) Moscow (USSR) Berlin (Germany)
Heavy Water Marker 	Southern Norway Western Ukraine (USSR) Dnepropetrovsk
Uranium Marker 	Great Plains (USA) Colorado Slovakia Jachymov Belgian Congo (Belgium) Shinkolobwe Ontario (Canada) Eldorado Mine



APPENDIX B: AXIS & ALLIES 1940 CONVERSION

If you want to use these rules with Axis & Allies 1940, the following changes will need to be made:

Home Nation: The Home Nation is well defined for players in Global War, however a definition is warranted for players of Global 1939 and Axis & Allies 1940. So for simplicity your home nation is;

Home Nation: Your nation's capitol, and each land zone that is contiguous with it you control at start of game. The home nation will never include conquered territories, colonies, or islands.

2.2 Transporting Uranium and Heavy Water: If you possess a territory with Uranium marker or Heavy Water marker you automatically have access to each component and can stockpile one of each per turn.

2.3 Research: You will need to succeed in 4 successful research rolls to develop all the necessary know-how and components to build a bomb. Each research roll costs 5 IPP and is successful on a “6”

3.2-3.4 Attacking with the Bomb: A bomb rolls 4D6, hitting on a 1-4 or does 6D6 strategic damage. Ignore reference to air superiority combat.

4.0 B-29 Bomber: The B-29 is an American bomber that can only be built once the player has developed both Long Range Aircraft and Heavy Bombers.

Atomic Bomb

ATOMIC BOMB	Stage 1	Stage 2	Stage 3	Complete
	6	6	6	6

B-29

Unit	Attack	Defense	Move	Cost
B-29	5	2	8	15



6.0 Add the following to Axis & Allies 1940 Set Ups;

Unit	Location
Research Markers 	Eastern United States (USA) Western United States (USA) Russia (USSR) Moscow Germany (Germany) Berlin
Heavy Water Marker 	Norway Telemark Ukraine (USSR) Dnepropetrovsk
Uranium Marker 	Western United States (USA) Colorado Slovakia Hungary (German) Jachymov Belgian Congo (UK) Shinkolobwe Yukon Territory (UK) Eldorado Mine

7.3 Carpet Bombing: Carpet bombing is an attack available only to Strategic bombers. Strategic bombers (and heavy strategic bombers) may attack forces in a land zone rolling 3 dice and getting a hit on each dice on a "1" (5 dice for heavy bombers). The attacking bomber is subject to air superiority combat and AA fire from AA guns (not facility guns) as if it were strategic bombing. No other land forces may fire back. Combat lasts 1 round after which the bomber must retreat.

Additional like pieces in this set can be purchased at Historical Board Gaming.

Nuclear Blast Marker

Uranium Marker

Research Marker

Nuclear Reactor Marker

Heavy Water Marker

Radiation Marker

If you have and corrections to this document or ideas to make it a better set please email us at info@historicalboardgaming.com.





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